



Cleared for Takeoff?

A Visual Analysis of U.S. Domestic Flight Delay Patterns (2019–2023)

Bobby Bagley | April 2026

Dataset: Flight Delay & Cancellation Dataset 2019–2023 · Bureau of Transportation Statistics via Kaggle

Introduction — About This Project

What are we analyzing?

Five years of U.S. domestic flight records from the Bureau of Transportation Statistics reveal when, where, why, and for whom delays happen — turning raw data into a practical guide for travelers, airlines, and analysts.

5M+

Total flights

5 yrs

2019–2023

237

Airports
mapped

18

Airlines

Story Structure

Chapter 1

Temporal Patterns — WHEN?

Chapter 2

Geographic Patterns — WHERE?

Chapter 3

Delay Causes — WHY?

Chapter 4

Airlines & Routes — WHO?

Introduction — About This Project

Dataset: Flight Delay and Cancellation Dataset (2019-2023)

<https://www.kaggle.com/datasets/patrickzel/flight-delay-and-cancellation-dataset-2019-2023>

Original data sourced from the U.S. Department of Transportation, Bureau of Transportation Statistics <https://www.transtats.bts.gov>

32

Features

29M

Rows

75.7K

Views on
Kaggle

17.8K

Downloads

Story Structure

Chapter 1

Temporal Patterns — WHEN?

Chapter 2

Geographic Patterns — WHERE?

Chapter 3

Delay Causes — WHY?

Chapter 4

Airlines & Routes — WHO?



Chapter 1 — Temporal Patterns

When do delays peak?

1.1 Monthly Delay Rate Heatmap (2019–2023)

Chart type: Seaborn Heatmap · Two dimensions at once: month and year



Summer Peak

June–August consistently show the highest delay rates, driven by peak leisure travel and school breaks.



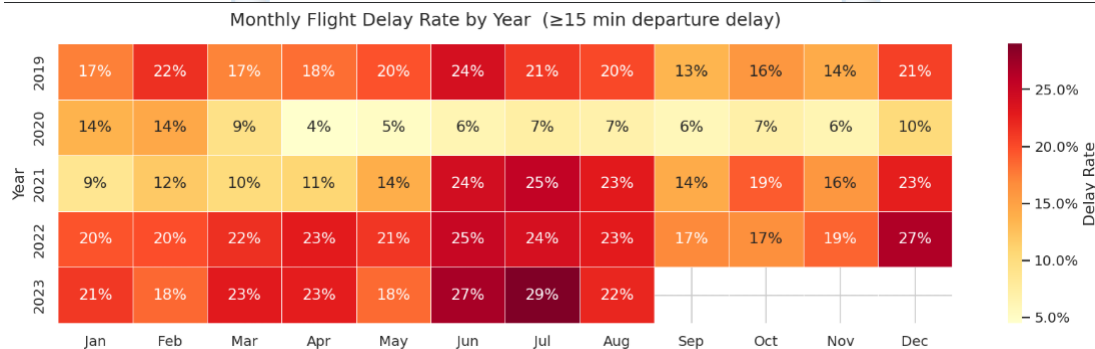
Holiday Surge

November–January heat up with Thanksgiving, Christmas, and New Year's holiday demand.



COVID Collapse

2020 shows a dramatic drop starting in March — the month COVID-19 was declared a global pandemic.



Key insight: COVID-19 is the single greatest structural break in five years of data — delays fell to near zero as air travel collapsed by over 60% in spring 2020.

1.2 Delay Rate by Hour of Day

Chart type: Altair Line Chart with Confidence Band · Each point = avg delay rate at that departure hour

7.3%

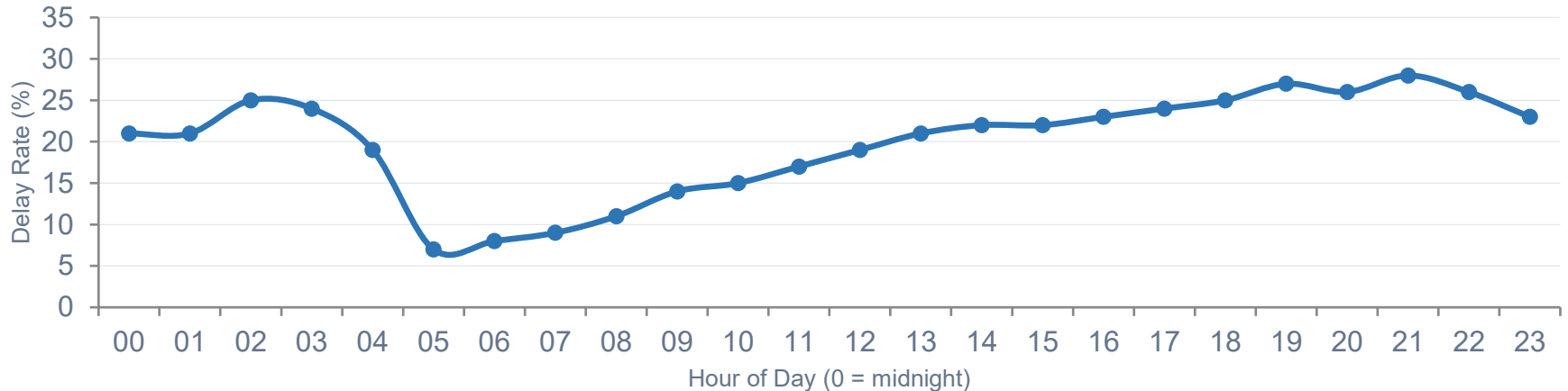
Lowest delay rate
0500 departure

28.3%

Highest delay rate
2100 departure

+21%

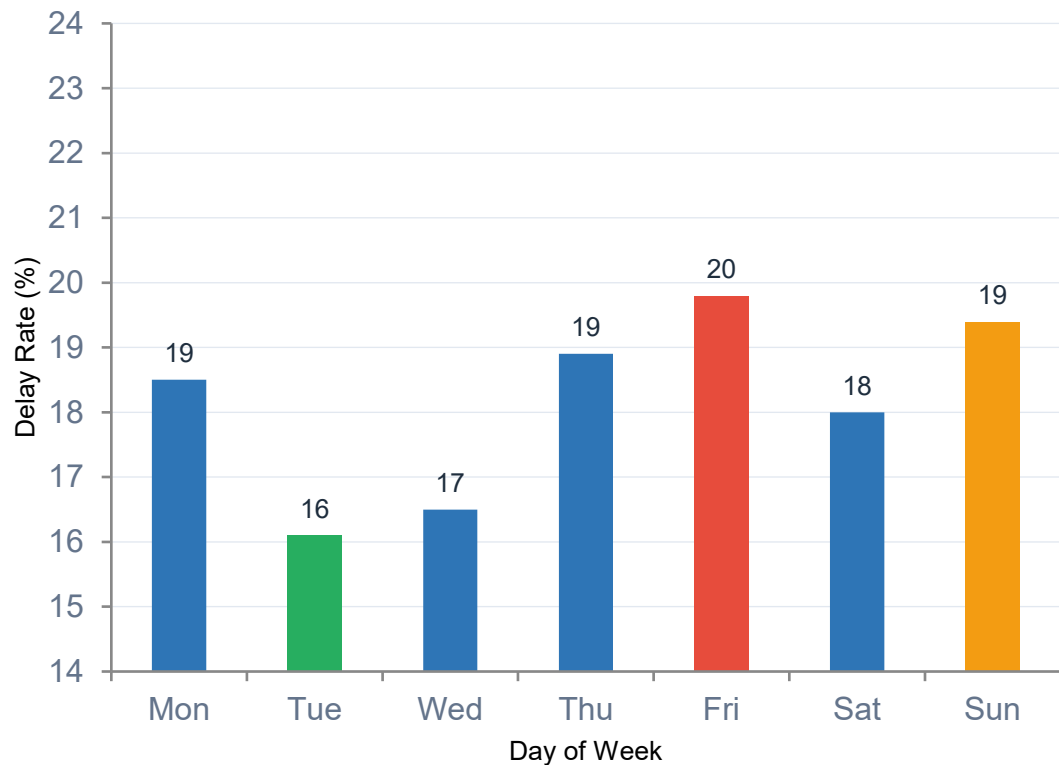
Increase from best
to worst hour



Domino Effect: Early delays compound throughout the day. A delay at 7AM cascades into delayed flights all afternoon.

1.3 Delay Rate by Day of Week

Chart type: Seaborn Bar Chart · 7 categories — ranking communicated immediately



Best Day

Tuesday — 16.1%
Lowest delay rate of the week. Mid-week travel is your safest bet.

Worst Day

Friday — 19.8%
End-of-workweek traffic pushes delay risk to its weekly peak.

Only a 3.7% spread separates best from worst — day of week is a far weaker predictor than time of day or month.



Chapter 1 Summary

Temporal patterns in flight delays are anything but random. At the monthly level, two seasonal peaks dominate the five-year record: summer (June–August), driven by peak leisure travel, and the winter holiday corridor (November–January). The most striking feature, however, is 2020 — where delay rates collapse nearly to zero in March as COVID-19 is declared a global pandemic.

Zooming into the hour of day, delays follow a strong positive trend beginning at 5:00 AM (7.3% rate) and climbing steadily to 9:00 PM (28.3% rate), consistent with a compounding domino effect where early delays propagate downstream throughout the day.

At the day-of-week level, the story is quiet: only a 3.7 percentage point spread separates Tuesday (best) from Friday (worst), confirming that time of day and time of year are far stronger predictors of delay risk than day of week alone.

Chapter 1 Key Takeaway: Fly early in the day, avoid summer and the holiday window — and never underestimate the domino effect.



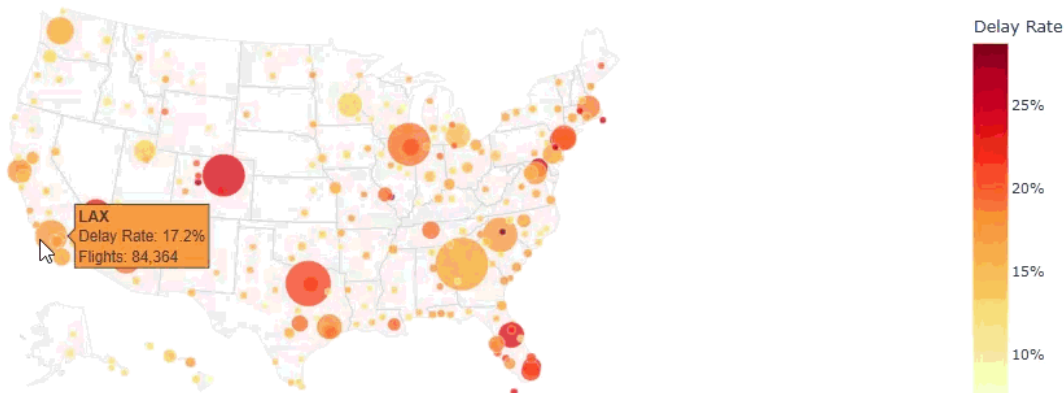
Chapter 2 — Geographic Patterns

Where are delays most severe?

2.1 Average Departure Delay by Origin Airport

Chart type: Plotly Bubble Map · Bubble size = flight volume · Color = delay rate (yellow → red)

Departure Delay Rate by Origin Airport (2019–2023)
Bubble size = flight volume | Color = delay rate (yellow = low, red = high)



Key Finding

- Airport size does NOT predict delay rate
- High-traffic hubs and small regional airports perform similarly
- Operational efficiency matters more than volume

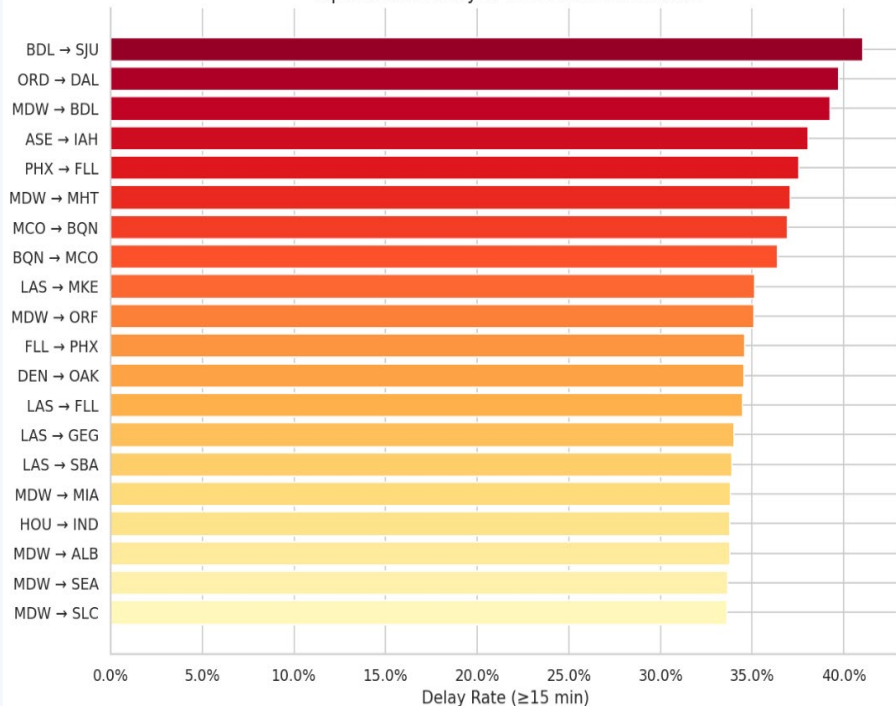
What to look for

- No strong correlation between bubble size and bubble color
- Delay risk appears geographically dispersed — not concentrated in one region

2.2 Top 20 Most-Delayed Routes

Chart type: Horizontal Bar Chart · Ranked by delay rate

Top 20 Most-Delayed U.S. Domestic Routes



Las Vegas (LAS) is the #2 contributor with 4 appearances on the most-delayed list.

Chicago Midway (MDW)

7 of 20
most-delayed routes
originate from MDW

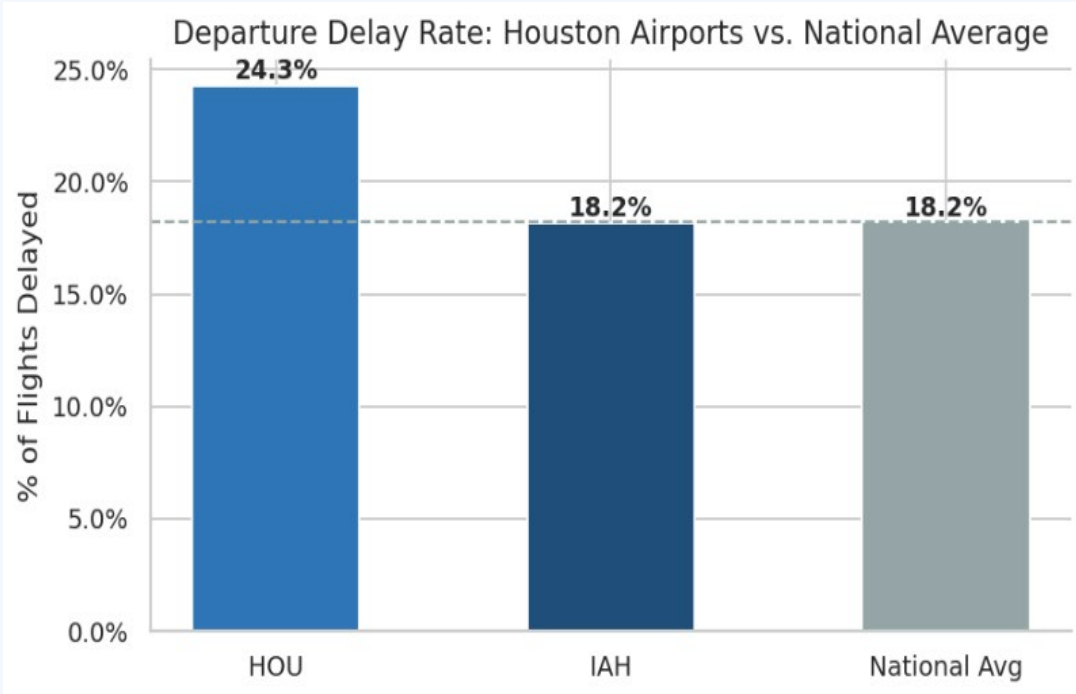
That's nearly double the contribution of any other single airport on the list.

Top 5 Most-Delayed Routes

- 1 BDL → SJU**
Hartford → San Juan, PR
- 2 ORD → DAL**
Chicago O'Hare → Dallas Love
- 3 MDW → BDL**
Chicago Midway → Hartford
- 4 ASE → IAH**
Aspen → Houston Bush
- 5 PHX → FLL**
Phoenix → Fort Lauderdale

2.3 Houston Spotlight — IAH vs. HOU vs. National Average

Chart type: Annotated Bar Chart · Local benchmark against national average



 **HOU — Hobby Airport**

24.3%
Delay Rate

One of the worst-performing airports in the country.

 **IAH — Bush Intercontinental**

18.2%
Delay Rate

Right at the national average — the lower-risk choice for Houston travelers.



Chapter 2 Summary

Geography shapes delay risk in ways that defy simple assumptions. Across 237 mapped U.S. airports, the most immediate takeaway is that airport size is not a reliable predictor of delay rate — large high-traffic hubs and smaller regional airports often perform similarly, suggesting that operational efficiency and geographic exposure matter far more than volume alone.

When zooming into the route level, a clear concentration emerges: Chicago Midway (MDW) alone accounts for 7 of the 20 most-delayed routes in the country, with Las Vegas (LAS) a distant second at four appearances. Delay risk clusters around specific origin airports with systemic operational challenges rather than being evenly distributed across the network.

For Houston travelers, this plays out in a particularly striking local example — Hobby Airport (HOU) posts a 24.3% delay rate, one of the worst in the country, while Bush Intercontinental (IAH) sits right at the national average of 18.2%.

Chapter 2 Key Takeaway: Where you fly from matters as much as when you fly — and not always in the direction you'd expect.

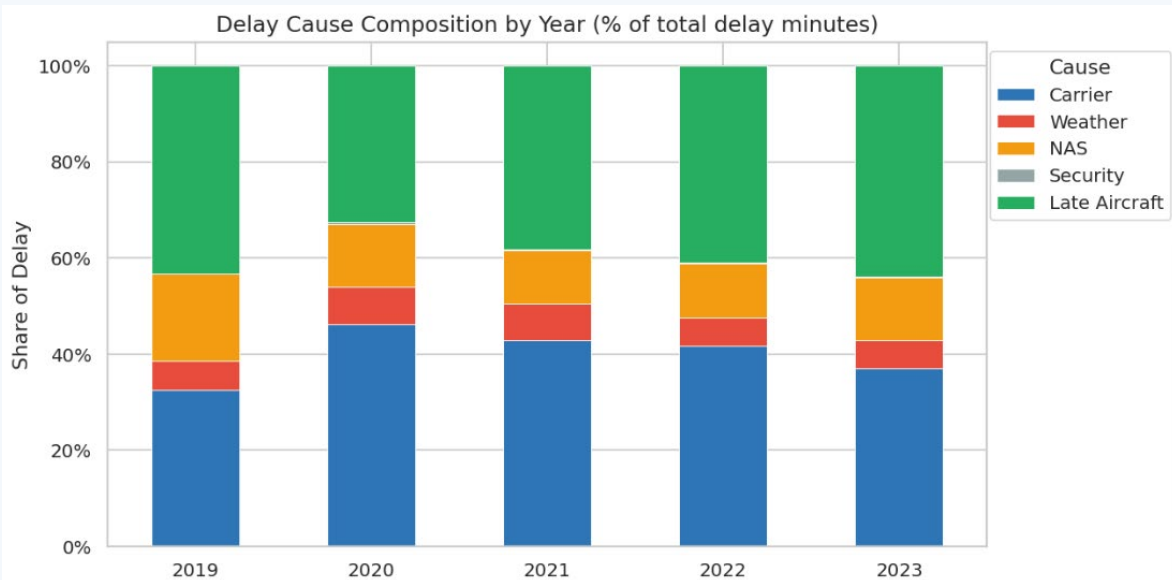


Chapter 3 — Delay Cause Breakdown

Why do flights get delayed?

3.1 Delay Cause Composition by Year (2019–2023)

Chart type: 100% Stacked Bar · Normalizes for volume — compares the mix of causes across years



#1 & #2 Causes

Carrier + Late Aircraft

Both 100% within airline control

Surprising finding

**Weather is nearly
the least common**

Ranks last (excl. Security)

Security delays

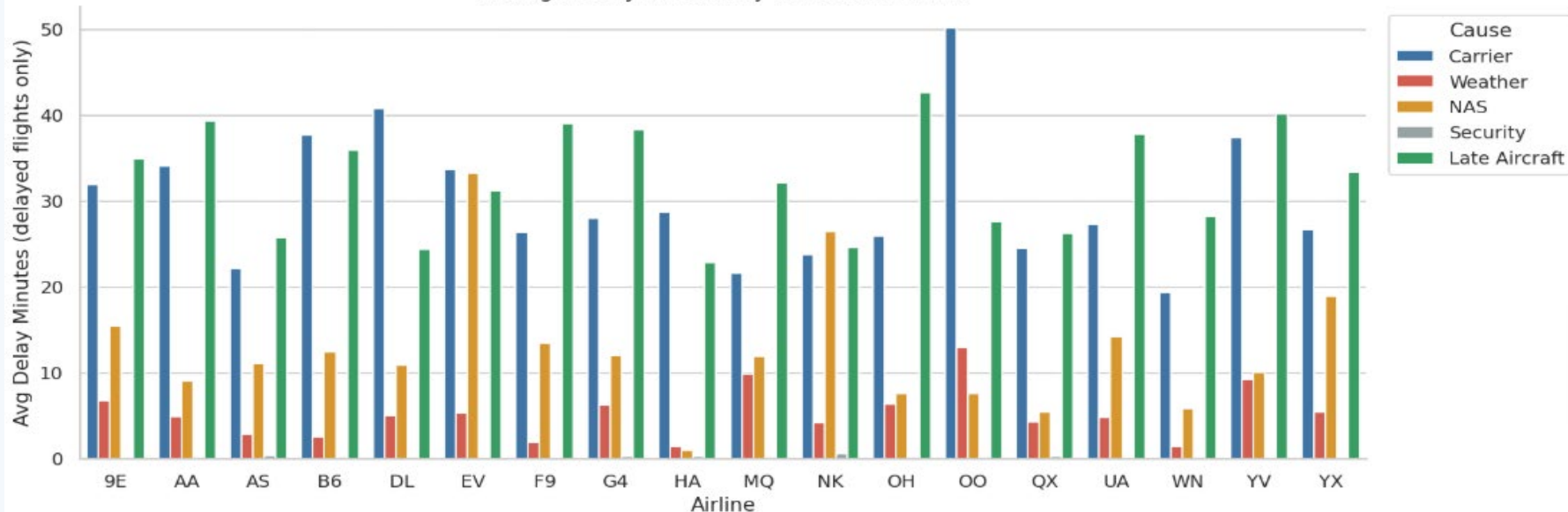
< 1–2%

Essentially negligible in all years

3.2 Average Delay Minutes by Cause and Airline

Chart type: Grouped Bar Chart · Delayed flights only · Minutes per delay cause per carrier

Average Delay Minutes by Cause and Airline



💡 Bottom Line: If an airline wants to meaningfully reduce delay severity, late aircraft turnaround efficiency is the single highest-leverage intervention — it dwarfs every other cause in both frequency and duration.

3.2 Average Delay Minutes by Cause and Airline

Chart type: Grouped Bar Chart · Delayed flights only · Minutes per delay cause per carrier

Late Aircraft — The Dominant Cause

- Late Aircraft delays are the #1 cause for virtually every airline
- Occurs when an incoming plane arrives late — leaving no aircraft for the next departure
- Controllable: it's a scheduling & fleet management failure, not an act of nature

Weather & Security — Minimal Impact

- Weather (red) contributes surprisingly little to delay length
- Security delays are so small they barely register on the chart
- Public perception is wrong — security is almost never the culprit

OO (SkyWest) — Standout Outlier

- SkyWest's Carrier delay towers above all others at ~50 minutes
- Operates for Delta, United, and American — scheduling pressure inherited from mainlines
- Not purely SkyWest's fault — a downstream effect of mainline decisions

NAS — Varies by Carrier

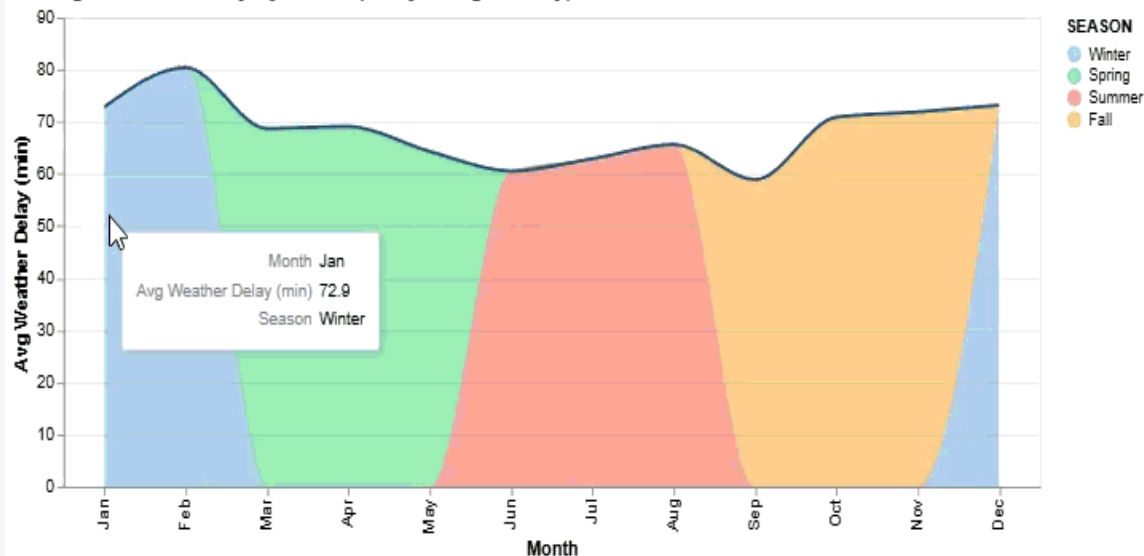
- Air traffic control delays vary meaningfully across airlines
- EV (ExpressJet) and G4 (Allegiant) show elevated NAS figures
- Reflects heavy reliance on congested regional & leisure-destination airports

 **Bottom Line:** If an airline wants to meaningfully reduce delay severity, late aircraft turnaround efficiency is the single highest-leverage intervention — it dwarfs every other cause in both frequency and duration.

3.3 Average Weather Delay by Month (Delayed flights only)

Chart type: Altair Layered Area Chart · Delayed flights only (≥ 15 min) · Color = season

Average Weather Delay by Month (delayed flights only)



❄ Winter (Jan & Dec)

~73 min

Snowstorms, ice, freezing temps

☀ Summer (Jun–Aug)

~70 min

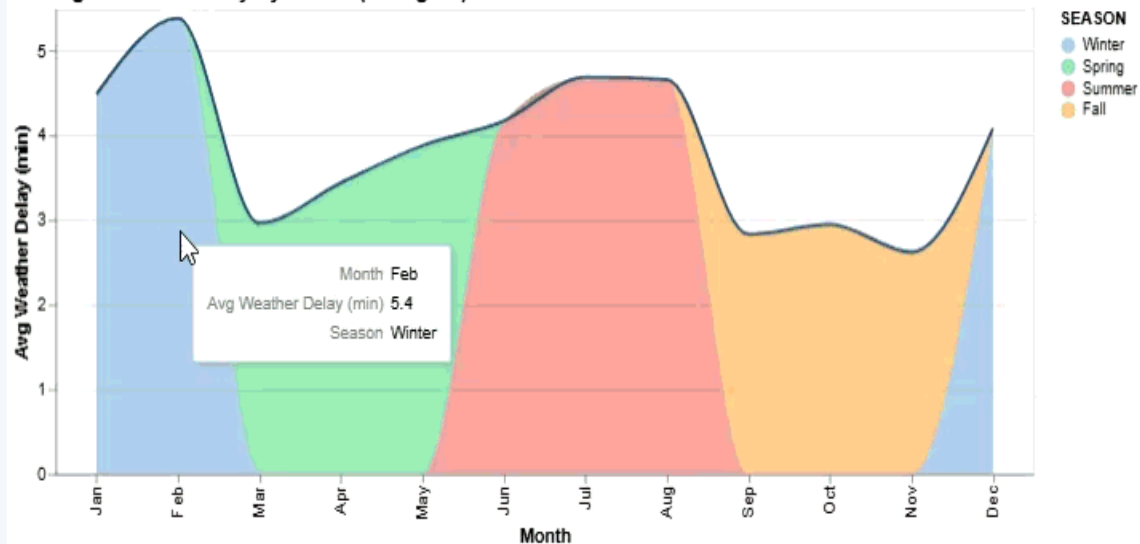
Afternoon thunderstorms in SE/MW

🔍 Surprise: Weather delays show a near-uniform pattern all year — no single dramatic spike. Both winter AND summer are dangerous. Summer thunderstorms are just as disruptive as winter snowstorms.

3.3 Average Weather Delay by Month (Delayed flights only)

Chart type: Altair Layered Area Chart · Delayed flights only (≥ 15 min) · Color = season

Average Weather Delay by Month (all flights)



❄ Winter (Jan & Dec)

~4.6 min

Snowstorms, ice, freezing temps

☀ Summer (Jun–Aug)

~4.5 min

Afternoon thunderstorms in SE/MW

~ : Here delays show a bimodal pattern peaking during the summer and winter. The peaks are more defined than the previous chart.



Chapter 3 Summary

The data delivers a result that cuts against conventional wisdom: the weather is not the primary culprit. Across all five years, Carrier and Late Aircraft delays dominate the composition chart, while Weather ranks as the least common cause — ahead of only Security. The reality is that most delays stem from circumstances entirely within the airline's control: maintenance issues, crew problems, fueling, baggage loading, and other operational failures.

Drilling into the airline-level breakdown reinforces this finding. Late Aircraft delays are the dominant cause for virtually every carrier — the most preventable of all. When an incoming aircraft arrives late, every subsequent flight assigned to that plane absorbs the delay in a chain reaction. The most striking individual data point belongs to SkyWest (OO), whose carrier delay towers above all others at ~50 minutes.

Chapter 3 Key Takeaway: Most delays are self-inflicted — airlines have the tools to fix most of their delay problem.

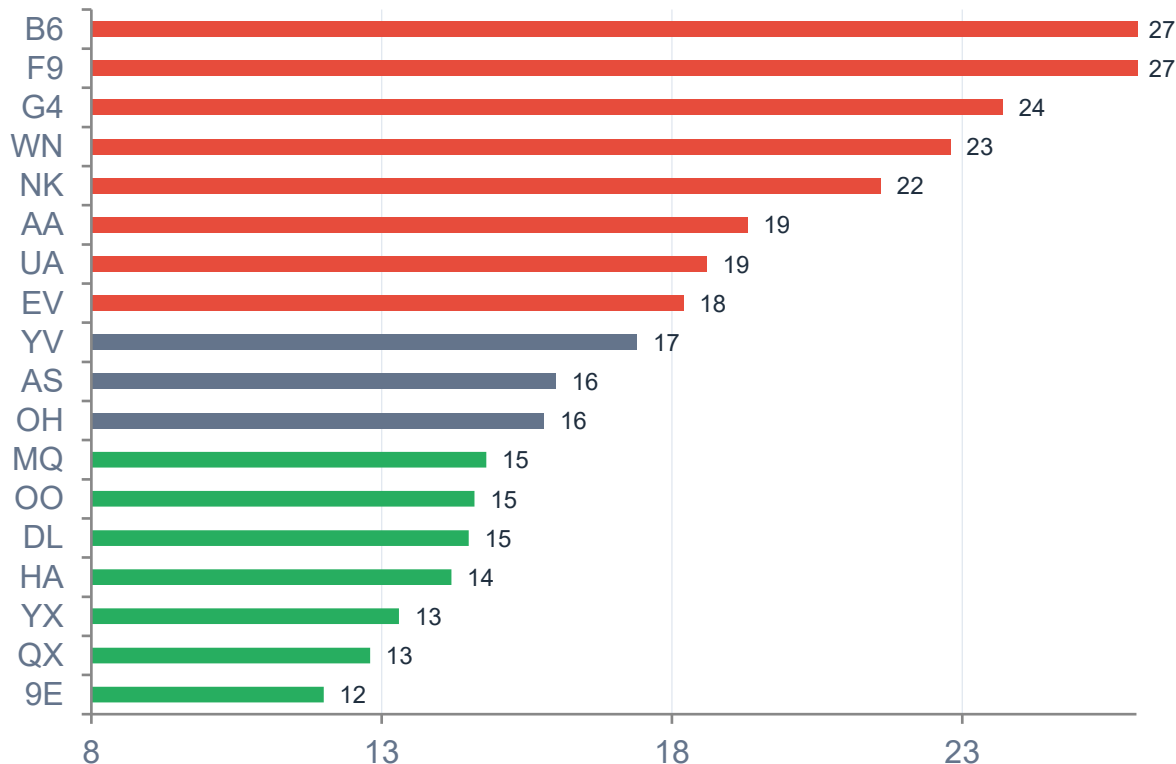


Chapter 4 — Airline & Route Analysis

Who is most affected?

4.1 On-Time Report Card by Airline

Chart type: Ranked Horizontal Bar · Color = performance vs. national average (green = better, red = worse)



Low-cost carriers dominate the worst performers — you may get what you pay for.



Best Performer

**Endeavor Air
(9E) — ~12%**

Half the delay rate of the worst performers

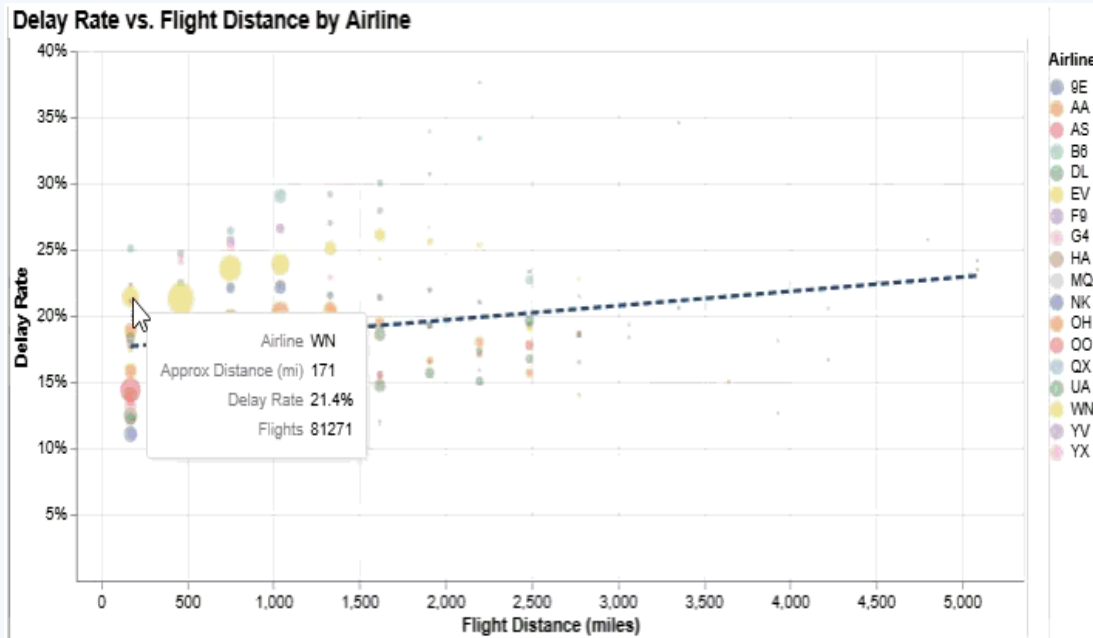


Top 5 Worst

- 1. JetBlue Airways**
- 2. Frontier Airlines**
- 3. Allegiant Air**
- 4. Southwest Airlines**
- 5. Spirit Airlines**

4.2 Delay Rate vs. Flight Distance by Airline

Chart type: Altair Scatter Plot with Regression Trend Line · Each point = an airline at a given distance bucket



What the trend shows

- Very weak positive relationship between distance and delay
- Longer flights do not meaningfully delay more than shorter ones
- The regression line is nearly flat — distance is a poor predictor

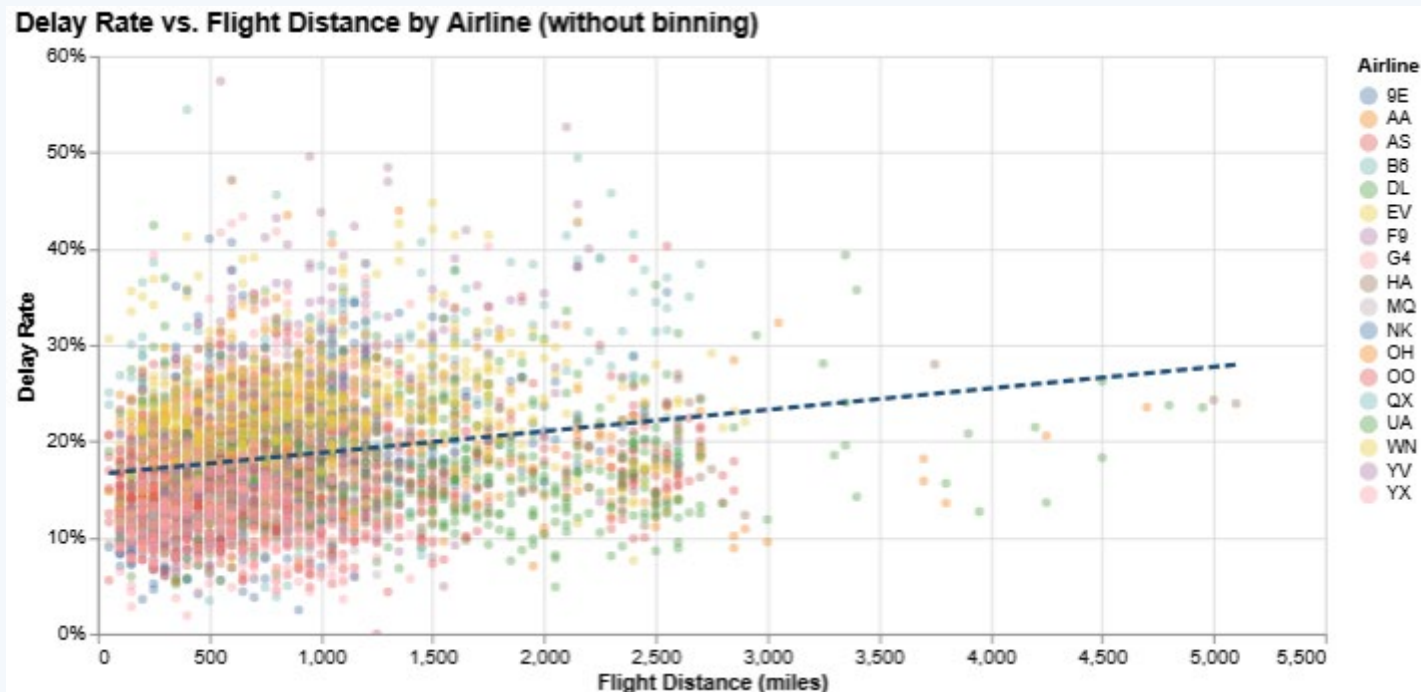
What actually predicts delays

- Time of day (Chapter 1) — strongest temporal predictor
- Month / season (Chapter 1) — summer & holidays
- Airline choice (Chapter 4) — up to 13% spread between best/worst
- Origin airport (Chapter 2) — MDW, LAS systemically worse

Practical insight: Whether you're flying 300 or 3,000 miles, your delay risk is driven far more by **WHEN** you fly and **WHO** you fly with than **HOW FAR**.

4.2 Delay Rate vs. Flight Distance by Airline (no binning)

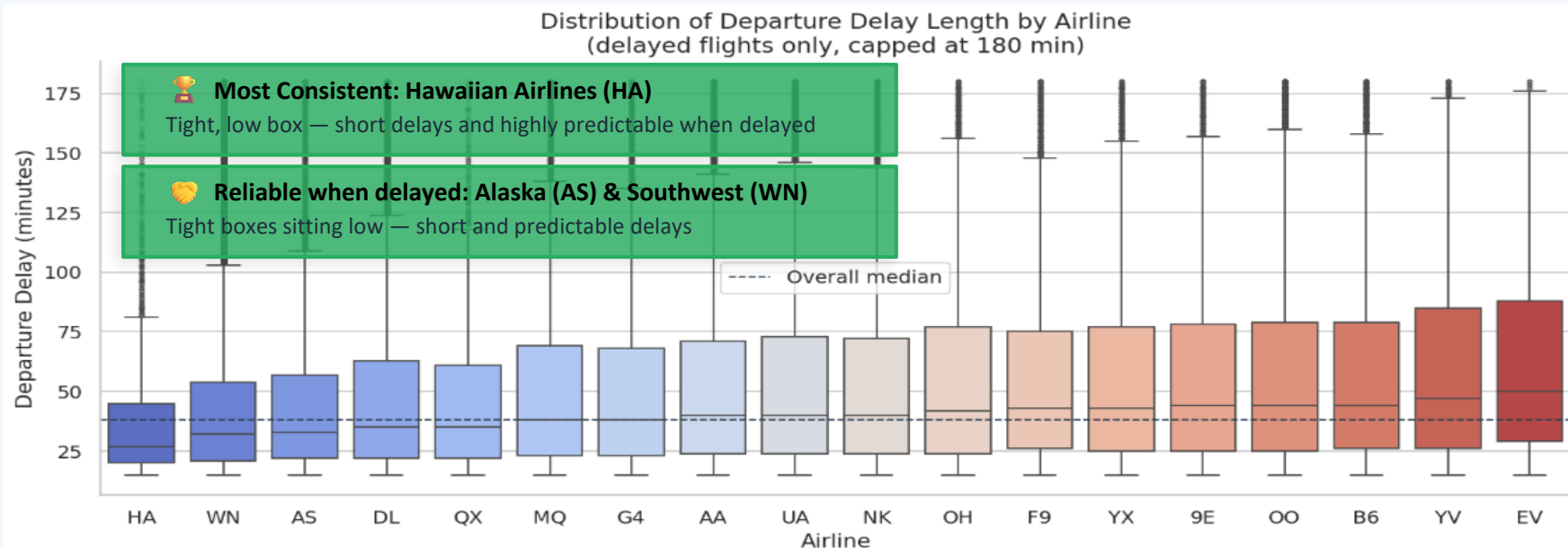
Chart type: Altair Scatter Plot with Regression Trend Line · Each point = an airline at a given distance bucket



Practical insight: Whether you're flying 300 or 3,000 miles, your delay risk is driven far more by **WHEN** you fly and **WHO** you fly with than **HOW FAR**.

4.3 Distribution of Delay Length by Airline

Chart type: Seaborn Box Plot · Delayed flights only, capped at 180 min · Sorted by median (left = best)



! Worst Performers: ExpressJet (EV) & Mesa (YV)

Wide boxes sitting high = they delay LONGER and MORE UNPREDICTABLY.
A delayed flight on EV or YV is a gamble — you never know how long you'll really wait.

Most airlines share a similar median (~38 min) — what separates good from bad is CONSISTENCY, not just average length.



Chapter 4 Summary

Not all airlines are created equal — and the data suggests price may be a reasonable proxy for punctuality. The five worst-performing carriers by delay rate are JetBlue, Frontier, Allegiant, Southwest, and Spirit — all fully independent low-cost or ultra-low-cost carriers. Their elevated delay rates reflect the same aggressive turnaround times and high aircraft utilization that keep ticket prices low, making them disproportionately vulnerable to the compounding domino effect identified in Chapter 1.

Flight distance, by contrast, turns out to be a poor predictor of delay risk. The scatter plot reveals only a very weak positive relationship between route distance and delay rate — whether you're flying a short regional hop or a cross-country haul, your odds of being delayed are driven far more by the airline you chose and the time you depart.

Where carriers diverge most sharply is not how often they delay, but how long and how consistently. Most airlines cluster near the same median of roughly 38 minutes — but Hawaiian (HA) stands out with a tight, low distribution, while ExpressJet (EV) and Mesa (YV) are not only the worst median performers but the most unpredictable. A delayed flight on those carriers is a genuine gamble.

Chapter 4 Key Takeaway: When it comes to flying, you may get what you pay for — choose wisely.



Key Takeaways

1

Fly early, fly mid-week

5AM departures have a 7.3% delay rate vs. 28.3% at 9PM. Tuesday is the safest day, Friday the riskiest.

2

Avoid summer & holidays

June–August and November–January are the peak delay seasons. COVID proved how structural these patterns are.

3

Size ≠ reliability

Big airports are not necessarily worse. Chicago Midway (MDW) dominates the worst-routes list despite being smaller than O'Hare.

4

Weather is not the villain

Carrier and Late Aircraft delays dominate — both 100% within airline control. Weather ranks last among major causes.

5

You get what you pay for

Low-cost carriers dominate the worst-performers list. Hawaiian Airlines has nearly half the delay rate of the worst carriers.

Bonus

Avoid Houston Hobby

For all of us in Houston: we should try to avoid flying out of HOU as it has one of the worst delay rate in the US. Drive an extra 10-30min north to IAH for a more reliable departure 😊